

Miles Jennings

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EDUCATION

University of California Irvine

Irvine, CA

Bachelor of Science in Computer Engineering

Sept. 2022 – June 2026

Relevant Coursework: Organization of Digital Computers, Electronics I–II, Network Analysis I–II, Data Structures and Algorithms, Discrete and Continuous-Time Signals, Digital Logic, Computer Architecture, VLSI, Embedded Systems, Digital Signal Processing, Analog IC Design

Organizations: HyperXite, FUSION, FASAE

EXPERIENCE

P2S Technology Intern

Long Beach, California

P2S Engineering

June 2025 – Present

- Assisted in the design and documentation of low-voltage and IT infrastructure systems for large scale buildings
- Supported engineers with **CAD** layout updates, wiring diagrams, and equipment schedules using AutoCAD and Revit
- Collaborated with multidisciplinary teams to coordinate technology system requirements with electrical and mechanical designs

HyperXite – SpaceX Hyperloop

Irvine, California

Control Systems Engineer

Aug. 2025 – Present

- Designed, tested, and optimized the Hyperloop pod's control system hardware using an STM32H753ZI microcontroller programmed in **C/C++**, interfacing with thermistors, time-of-flight sensors, and INA219 current sensors while evaluating components for performance and cost improvements over prior pod iterations
- Ensured hardware–software compatibility by defining system constraints and key telemetry parameters based on feedback from team surveys identifying the most critical data points to track in the GUI
- Implemented a Continuous Integration (CI) pipeline using GitHub Actions to streamline testing, version control, and code deployment across subteam

PROJECTS

AudioVisor | C, PDM, nRF52840, TypeScript, Vite, Supabase

Sept. 2025 – Present

- Built an audio-reactive visor using an nRF52840, utilizing **C** to receive and handle digital microphone input through **PDM** communication for real-time sound magnitude/direction detection
- Developed a real-time web dashboard for the AudioVisor system to visualize and analyze audio data, achieving **84%** accuracy in detecting sound direction and amplitude during initial trials
- Researched and evaluated components to satisfy sensitivity, latency, and power needs, guiding system architecture and part selection

Hand Gesture Rover | C, NRF, ICM

April 2024

- Designed and built a gesture-controlled rover using Arduino and NRF24L01 modules for wireless communication
- Programmed real-time motion tracking with the ICM's 9-axis IMU to convert hand tilts into directional movement, applying data filtering and SPI-based communication
- Designed and assembled both transmitter and rover hardware, including power circuitry, motor driver integration, and sensor mounting for reliable embedded control

Building Management System | Python, Raspberry Pi, LCD, DHT-11

April 2025 - June 2025

- Designed and constructed a fully functional Building Management System (BMS) on Raspberry Pi using real-time data from DHT-11 and PIR sensors to control HVAC, ambient lighting, fire alarms, and security systems
- Programmed an interactive LCD interface and logging system to display live status updates, trigger event-based alerts, and ensure time-stamped logs with synchronization and interrupt-driven input handling

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Verilog, RISC-V, MIPS Assembly, MATLAB

Software/Tools: Revit, AutoCAD, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm

Hardware: Oscilloscope, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies